

## QUALIFYING EXAMINATION, Part 2

### Solutions

#### Problem 1: Electromagnetism I

From symmetry considerations, the electric field is in all cases in the radial direction  $\hat{r}$  with magnitude  $E = E(r)$  so in general  $\vec{E} = E(r)\hat{r}$ . Applying Gauss's law to a sphere of radius  $r$ , we have

$$4\pi r^2 E(r) = \frac{Q_{\text{enc}}}{\epsilon_0} ,$$

or

$$E = \frac{Q_{\text{enc}}}{4\pi\epsilon_0} \frac{1}{r^2} ,$$

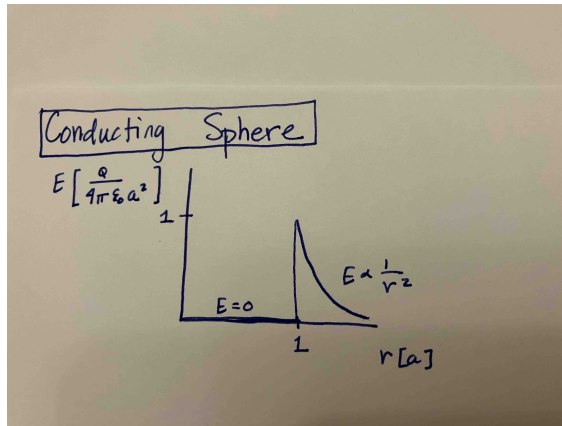
where  $Q_{\text{enc}}$  is the charge enclosed inside the sphere of radius  $r$ .

In all cases, for  $r > a$  the enclosed charge is  $Q_{\text{enc}} = Q$  and therefore the electric field outside is

$$E_{\text{outside}}(r) = \frac{Q}{4\pi\epsilon_0} \frac{1}{r^2} .$$

Next we determine the electric field inside the sphere:

(a) For a conducting sphere,  $Q_{\text{enc}} = 0$  for any  $r < a$  and  $E_{\text{inside}}(r) = 0$ .

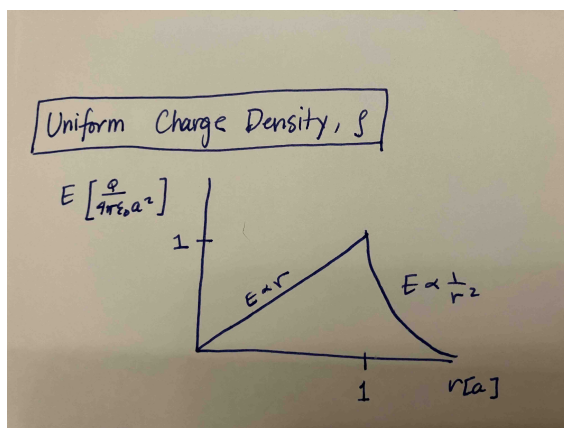


(b) For a uniform charge density, we have

$$Q_{\text{enc}} = \left( \frac{r}{a} \right)^3 Q ,$$

so that

$$E_{\text{inside}}(r) = \frac{Q}{4\pi\epsilon_0} \frac{r}{a^3} .$$



(c) The charge density is  $\rho = Ar^2$ . To determine the constant  $A$ , we use

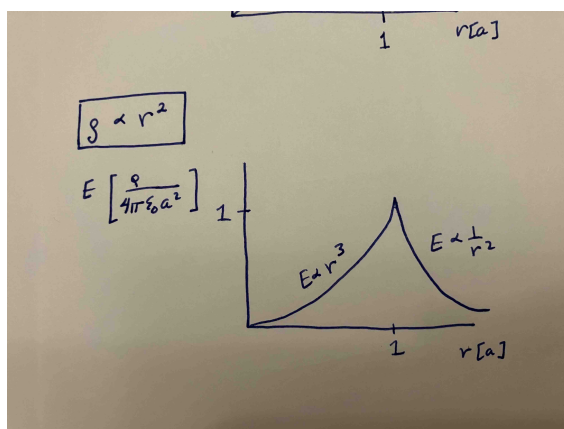
$$Q = 4\pi \int_0^a r^2 \rho(r) dr = 4\pi A \int_0^a r^4 dr = 4\pi A \frac{a^5}{5}$$

to find  $A = \frac{5}{4\pi a^5} Q$ . The enclosed charge inside a sphere of radius  $r < a$  is then

$$Q_{\text{enc}} = 4\pi \int_0^r r'^2 \rho(r') dr' = 4\pi A \frac{r^5}{5} = \frac{Q}{a^5} r^5,$$

and

$$E_{\text{inside}}(r) = \frac{Q}{4\pi\epsilon_0} \frac{r^3}{a^5}.$$



(d) The charge density is  $\rho = Ar^{-2}$ . We determine  $A$  from

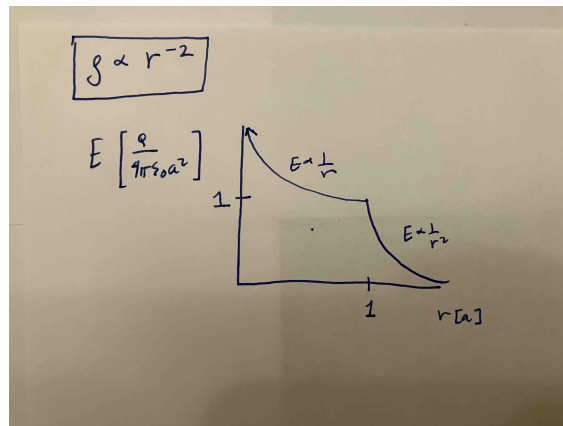
$$Q = 4\pi \int_0^a r^2 \rho(r) dr = 4\pi A \int_0^a dr = 4\pi Aa$$

and find  $A = \frac{Q}{4\pi a}$ . The enclosed charge inside a sphere of radius  $r < a$  is then

$$Q_{\text{enc}} = 4\pi \int_0^r r'^2 \rho(r') dr' = 4\pi Ar = \frac{Q}{a} r,$$

and

$$E_{\text{inside}}(r) = \frac{Q}{4\pi\epsilon_0 a} \frac{1}{r}.$$



## Problem 2: Electromagnetism II

(a) For the magnetostatic case, since the plane is uncharged the electric field is  $\vec{E} = 0$  everywhere.

Using the right-hand rule, the magnetic field points  $\vec{B}$  in the  $-\hat{y}$  direction for  $z > 0$ . Its magnitude can be determined by Ampere's law, using a rectangular Amperian loop of edge length  $l$  oriented in the  $y$  direction enclosing the current in the plane. We find

$$\begin{aligned} \int \vec{B} \cdot d\vec{l} &= \mu_0 I_{enc} \\ \Rightarrow 2lB &= \mu_0 K_0 l. \end{aligned}$$

We conclude that

$$\vec{B} = -\frac{\mu_0 K_0}{2} \hat{y}.$$

(b) For the time-dependent case, we still have  $\rho = 0$  so the scalar potential is  $\phi(\vec{x}, t) = 0$  everywhere.

For the vector potential, by translational symmetry it can depend only on  $z$  and  $t$ . At time  $t$  for a point at height  $z$  above the plane, then we evaluate the current density at retarded time  $t'$ . For an arbitrary point in the  $x$ - $y$  plane  $\vec{x}'$  that is at radius  $r$  from the origin, then  $|\vec{x} - \vec{x}'| = \sqrt{r^2 + z^2}$ . The current density is non-zero in a circular region where  $t' = t - \sqrt{r^2 + z^2}/c > 0$ , or  $r < \sqrt{(ct)^2 - z^2}$ . Thus, we can evaluate the integral as:

$$\begin{aligned} \vec{A}(z, t) &= \frac{\mu_0}{4\pi} \int_0^{2\pi} d\phi \int_0^{\sqrt{(ct)^2 - z^2}} \frac{(K_0 \hat{x}) r dr}{\sqrt{r^2 + z^2}} \\ &= \frac{\mu_0 K_0 \hat{x}}{2} \left[ \sqrt{(ct)^2 - z^2 + z^2} - \sqrt{z^2} \right], \end{aligned}$$

where we've used the form given above to evaluate the integral. The vector potential in the region  $z > 0$  is then given by:

$$\vec{A}(z, t) = \begin{cases} \frac{\mu_0 K_0}{2} (ct - z) \hat{x} & t > z/c \\ 0 & \text{otherwise} \end{cases}$$

(c) The electric field is given by  $\vec{E} = -\nabla\phi - \frac{\partial\vec{A}}{\partial t}$  so

$$\vec{E}(z, t) = \begin{cases} -\frac{\mu_0 K_0 c}{2} \hat{x} & t > z/c \\ 0 & \text{otherwise} \end{cases}$$

again considering the region  $z > 0$ . For the magnetic field  $\vec{B} = \nabla \times \vec{A}$ , which in this case has only one non-vanishing term:  $\vec{B} = \left(\frac{\partial A_x}{\partial z}\right) \hat{y}$ . So, we have:

$$\vec{B}(z, t) = \begin{cases} -\frac{\mu_0 K_0}{2} \hat{y} & t > z/c \\ 0 & \text{otherwise} \end{cases}$$

which matches the magnetostatic case at long times.

(d) For the magnetostatic case, since  $\vec{E} = 0$ , the Poynting vector is also  $\vec{S} = 0$  and no power is radiated (as expected in the static case).

(e) For the time dependent current, then we have:

$$\vec{S} = \frac{1}{\mu_0}(\vec{E} \times \vec{B}) \frac{1}{\mu_0} \left( -\frac{\mu_0 K_0 c}{2} \right) \left( -\frac{\mu_0 K_0}{2} \right) \hat{z} = \frac{\mu_0 K_0^2 c}{4} \hat{z} .$$

Thus the power passing through a plane of area  $A$  at height  $z > 0$  is

$$P(t) = \begin{cases} \frac{\mu_0 K_0^2 c A}{4} & t > z/c \\ 0 & \text{otherwise} \end{cases}$$